Algebra 2 Exponential & Log Cheat Sheet

Unit Focus: *exponential functions, e, and the logarithm*

1 KEY VOCABULARY

exponential function	A function with the variable in the exponent: $f(x) = a \cdot b^x$, with $a \neq 0$, $b > 0$, $b \neq 1$.
growth factor	$b > 1$. Output is multiplied by b each time x increases by 1. $b = 1 + r$.
decay factor	$0 < b < 1$. Output shrinks by the factor b each step. $b = 1 - r$.
asymptote	A line the graph approaches but never touches. For $y = a \cdot b^x$ it is $y = 0$.
e	The irrational number $e \approx 2.71828$, the base of continuous growth and of natural logs.
compound interest	Interest earned on the principal and on interest already earned.
logarithm	The exponent that a base must be raised to in order to get a number: $\log_b x = y$ means $b^y = x$.
common log, natural log	$\log x$ means $\log_{10} x$. $\ln x$ means $\log_e x$.
argument	The input of a log, x in $\log_b x$. It must be positive.
extraneous solution	A result from the algebra that fails in the original equation (for example, it makes a log argument ≤ 0).
change of base	$\log_b x = \frac{\ln x}{\ln b}$. Lets a calculator evaluate any base.

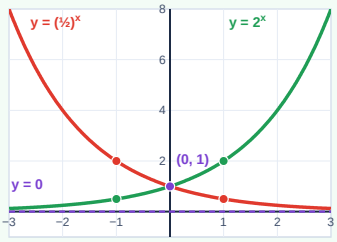
2 EXPONENTIAL GROWTH & DECAY

$$f(x) = a \cdot b^x$$

Growth: $b > 1$
rising; $b = 1 + r$
5% → b = **1.05**

Decay: $0 < b < 1$
falling; $b = 1 - r$
5% → b = **0.95**

- **Domain:** all real numbers
- **Range:** $y > 0$ (when $a > 0$)
- ***y*-intercept:** $(0, a)$
- **Horizontal asymptote:** $y = 0$
- No x-intercept. Passes through $(1, ab)$.



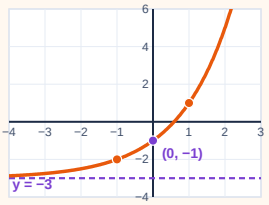
3 TRANSFORMATIONS & FEATURES

$$g(x) = a \cdot b^{x-h} + k$$

- h: shifts **right** h units
- k: shifts **up** k units, and the **asymptote moves to $y = k$**
- $a < 0$: reflects over the x-axis

Example: Graph $f(x) = 2^{x+1} - 3$

Shift left 1, down 3.
Asymptote: $y = -3$
y-int: $2^1 - 3 = -1$ → $(0, -1)$
Range: $y > -3$



4 THE NUMBER *E* & COMPOUND INTEREST

$e \approx 2.71828$ (the value e approaches as $(1 + \frac{1}{n})^n$ gets huge)

***n* times per year**	**Continuously**
$A = P(1 + \frac{r}{n})^{nt}$	$A = Pe^{rt}$

P = principal, ***r*** = rate as a decimal, ***t*** = years, ***n*** = periods per year

Example: $2,000 at 5% for 6 years

Quarterly ($n = 4$): $A = 2000(1 + \frac{0.05}{4})^{24} \approx$ **$2,694.70**

Continuous: $A = 2000e^{0.05(6)} = 2000e^{0.3} \approx$ **$2,699.72**

5 LOGARITHM DEFINITION & CONVERSION

$$\log_b x = y \iff b^y = x$$

$b > 0$, $b \neq 1$, and $x > 0$. The log **is** the exponent.

Exponential form	Logarithmic form
$2^5 = 32$	$\log_2 32 = 5$
$9^{\frac{1}{2}} = 3$	$\log_9 3 = \frac{1}{2}$
$10^{-2} = 0.01$	$\log 0.01 = -2$
$e^4 = x$	$\ln x = 4$

Base stays the base. Exponent and answer swap sides.

6 EVALUATING LOGS WITHOUT A CALCULATOR

Ask: **"b to what power gives x?"** Write x as a power of b.

$\log_4 64$: $4^3 = 64$, so **3**
$\log_2 \frac{1}{8}$: $2^{-3} = \frac{1}{8}$, so **−3**

$\log_{25} 5$: $25^{\frac{1}{2}} = 5$, so $\frac{1}{2}$
$\log 1000 =$ **3** $\ln e^5 =$ **5**

Special values:

$\log_b 1 = 0$ $\log_b b = 1$ $\log_b b^x = x$ $b^{\log_b x} = x$

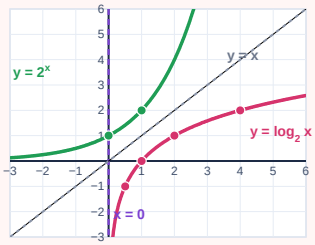
7 LOG GRAPH, DOMAIN & RANGE

$y = \log_b x$ ($b > 1$) is the **inverse** of $y = b^x$: reflect over $y = x$.

- **Domain:** $x > 0$
- **Range:** all real numbers
- **Vertical asymptote:** $x = 0$
- ***x*-intercept:** $(1, 0)$; also $(b, 1)$

Example: Domain of $f(x) = \log_3(2x - 6)$

$2x - 6 > 0$ → $x > 3$. Asymptote $x = 3$.



8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Write the rate as a decimal: 5% = 0.05.
- Growth factor is $1 + r$, decay is $1 - r$.
- Log arguments must be positive.
- The log is an exponent: $\log_2 8 = 3$.
- log means base 10; ln means base e.

COMMON MISTAKES:

- Using $b = 0.05$ instead of 1.05 for 5% growth.
- Dropping the base: writing log 8 for $\log_2 8$.
- Thinking $\log x$ can be negative or 0 inside: $\log(-3)$ is undefined.
- Calling e a variable. It is a constant, about 2.718.

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More Practice: *log properties, solving equations, and modeling*

1 PROPERTIES OF LOGARITHMS

For $b > 0$, $b \neq 1$, $M > 0$, $N > 0$:

Product	$\log_b(MN) = \log_b M + \log_b N$
Quotient	$\log_b(\frac{M}{N}) = \log_b M - \log_b N$
Power	$\log_b(M^p) = p \cdot \log_b M$

Example: $\log_2 8 + \log_2 4 = 3 + 2 = 5 = \log_2(8\cdot4) = \log_2 32$ ✓

× $\log(a + b) \neq \log a + \log b$. **There is no rule for a log of a sum.**

2 EXPANDING & CONDENSING

Expand $\log_2(\frac{8x^3}{y})$
$= \log_2 8 + \log_2 x^3 - \log_2 y$
$= \mathbf{3 + 3\log_2 x - \log_2 y}$

Condense $2 \ln x + \ln 5 - \ln y$
$= \ln x^2 + \ln 5 - \ln y$
$= \mathbf{\ln(\frac{5x^2}{y})}$

Evaluate by condensing: $\log_6 4 + \log_6 9 = \log_6(4\cdot9) = \log_6 36 = \mathbf{2}$

To condense: power rule first (move coefficients up), then add logs → multiply, subtract logs → divide.

3 CHANGE OF BASE

$$\log_b x = \frac{\log x}{\log b} = \frac{\ln x}{\ln b}$$

Use it when your calculator only has log and ln.

Example: Evaluate $\log_5 40$.

Step 1: $\log_5 40 = \frac{\ln 40}{\ln 5}$

Step 2: $\approx \frac{3.6889}{1.6094}$

Step 3: $\approx$ **2.2920** (round last)

Check: $5^{2.2920} \approx 40$ ✓

4 SOLVING EXPONENTIAL EQUATIONS

Same base: $4^{x+1} = 8^{x-1}$
$2^{2(x+1)} = 2^{3(x-1)}$
$2x + 2 = 3x - 3 \rightarrow \mathbf{x = 5}$
Check: $4^6 = 4096 = 8^4$ ✓

Use logs: $2\cdot5^x = 70$
$5^x = 35$ (isolate first)
$x = \frac{\log 35}{\log 5} \approx \mathbf{2.209}$
Check: $2\cdot5^{2.209} \approx 70$ ✓

Rewrite bases: $27^x = 9 \rightarrow 3^{3x} = 3^2 \rightarrow x = \frac{2}{3}$ $(27^{\frac{2}{3}} = 9$ ✓)

Base *e*: $e^{2x} = 9 \rightarrow 2x = \ln 9 \rightarrow x = \frac{\ln 9}{2} \approx \mathbf{1.099}$ $(e^{2(1.099)} \approx 9$ ✓)

5 SOLVING LOG EQUATIONS (CHECK!)

1. Condense to one log. **2.** Convert to exponential form. **3.** Solve. **4. CHECK** every answer in the original equation.

Example: Solve $\log_2 x + \log_2(x - 2) = 3$

$\log_2[x(x - 2)] = 3 \rightarrow x(x - 2) = 2^3 = 8$
$x^2 - 2x - 8 = 0 \rightarrow (x - 4)(x + 2) = 0 \rightarrow x = 4$ or $x = -2$

***x* = −2:** $\log_2(-2)$ is undefined. **Reject (extraneous).**

***x* = 4:** $\log_2 4 + \log_2 2 = 2 + 1 = 3$ ✓

$x = 4$ Also: $\log_3(2x + 1) = 2 \rightarrow 2x + 1 = 9 \rightarrow x = 4$ ✓. Same-base logs: $\log(x + 6) = \log(2x - 1) \rightarrow x = 7$ ✓

6 MODELING FORMULAS

Growth or decay	$y = a(1 \pm r)^t$
Compound interest	$A = P(1 + \frac{r}{n})^{nt}$
Continuous	$A = Pe^{rt}$ (decay: e^{-kt})
Half-life *h*	$N = N_0(\frac{1}{2})^{\frac{t}{h}}$
Doubling time *d*	$N = N_0\cdot2^{\frac{t}{d}}$; at continuous *r*: $t = \frac{\ln 2}{r}$
Acidity	$\text{pH} = -\log[\text{H}^+]$
Sound level	$L = 10\log(\frac{I}{I_0})$ dB

To solve for *t*: **isolate** the exponential part, then take log or ln of both sides. Example: $[\text{H}^+] = 3.2 \times 10^{-4}$ gives pH $= -\log(3.2 \times 10^{-4}) \approx$ **3.49**.

7 REAL-WORLD EXAMPLES

Example: A population follows $P = 500e^{0.04t}$. When does it reach 2000?

Step 1: $2000 = 500e^{0.04t} \rightarrow e^{0.04t} = 4$

Step 2: $0.04t = \ln 4 \rightarrow t = \frac{\ln 4}{0.04}$

Step 3: $t \approx$ 34.7 years

Doubling time at 4%: $\frac{\ln 2}{0.04} \approx 17.3$ years

Example: An 80 mg sample has a half-life of 6 days. When are 15 mg left?

$15 = 80(\frac{1}{2})^{\frac{t}{6}} \rightarrow (\frac{1}{2})^{\frac{t}{6}} = \frac{15}{80} = 0.1875$

$\frac{t}{6} = \frac{\ln 0.1875}{\ln 0.5} \rightarrow t = 6\cdot\frac{\ln 0.1875}{\ln 0.5} \approx$ 14.5 days

8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Isolate the exponential before you take a log.
- On a calculator, put parentheses around the top and the bottom of a log fraction.
- Keep full decimals until the final answer.
- Substitute every solution back into the original equation.
- Include units in modeling answers.

COMMON MISTAKES:

- $\log(a+b) = \log a + \log b$. No!
- $\frac{\log M}{\log N} \neq \log(\frac{M}{N})$; $(\log M)^2 \neq 2\log M$.
- Forgetting to reject extraneous roots.
- Rounding early (use 4+ decimals mid-problem).
- Using *r* instead of 1 + *r*.